3DAC: Learning Attribute Compression for Point Clouds

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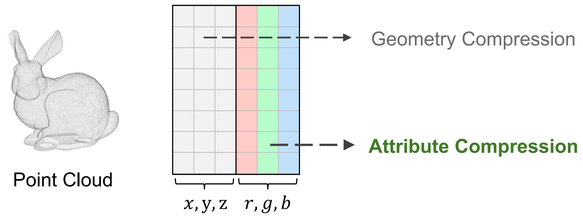
Introduction

Point Cloud Compression



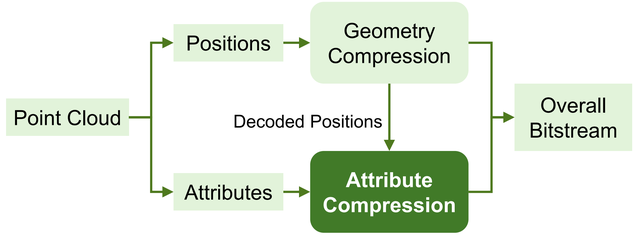
Contributions

- A learning-based framework for attribute compression.
- An attribute-oriented entropy model.
- State-of-the-art compression performance.
- Paper, code and project: https://fatpeter.github.io/

Problem Definition

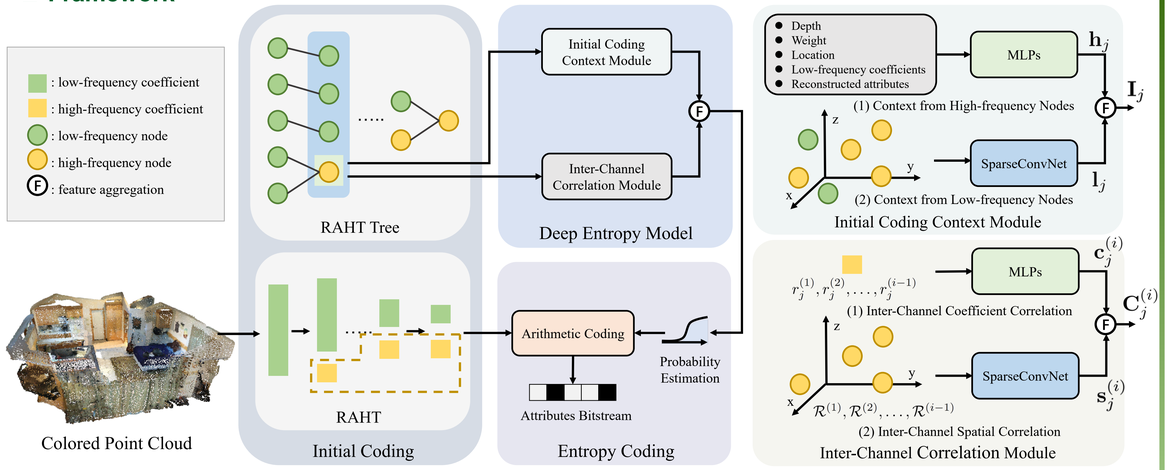
Goal: Given a 3D point cloud, its geometry is assumed to have been transmitted separately and we mainly focuses on the task of point cloud **attribute compression**.

Compression Pipeline



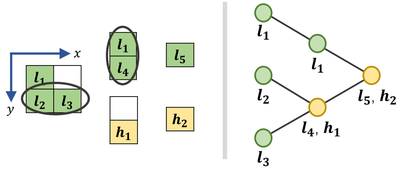
Method

Framework



Initial Coding

We adopt Region Adaptive Hierarchical Transform (**RAHT**) for initial coding and construct the **RAHT tree** for context feature extraction.



(a) RAHT (b) RAHT Tree

Deep Entropy Model

Formulation

Given transformed coefficients R, we factorize its probability distribution at the attribute channel i with initial coding context $\mathbf{I}$ and inter-channel correlation $\mathbf{C}^{(i)}$ as follows:

$$q(\mathcal{R}^{(i)}) = \prod_j q\left(r_j^{(i)} \mid \mathbf{I}_j, \mathbf{C}_j^{(i)}\right).$$

Context Modelling

By exploring context information from the RAHT tree and the previous encoded attributes, we model the initial coding context $\mathbf{I}_j$ and the inter-channel correlation $\mathbf{C}_j^{(i)}$ with our Initial Coding Context Module and Inter-Channel Correlation Module.

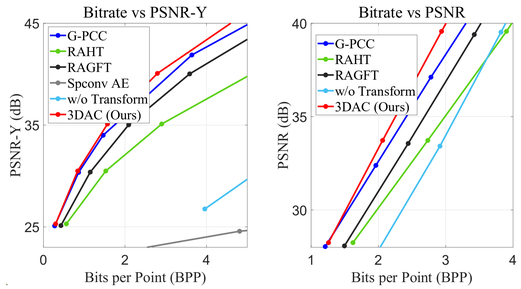
Learning

During training, we approximate the estimated distribution with the actual distribution by minimizing the cross-entropy loss:

$$\ell = -\sum_i \sum_j \log q(r_j^{(i)} \mid \mathbf{I}_j, \mathbf{C}_j^{(i)}).$$

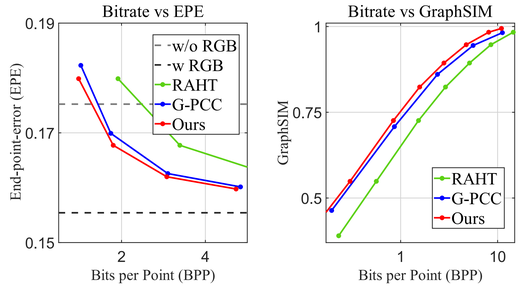
Experiments

Quantitative Results



(a) ScanNet (b) SemanticKITTI

Downstream Tasks



(a) Scene Flow Estimation (b) Quality Assessment

Qualitative Results

BPP: 3.46 $PSNR_y$: 34.63 (a) RAHT

BPP: 2.11 $PSNR_y$: 34.60 (b) G-PCC

BPP: 2.07 $PSNR_y$: 34.63 (c) Ours